Heart Disease Analysis

1. PROJECT TITLE

Heart Disease Analysis using SQL, Tableau Dashboard, and Flask Web Integration

2. PROJECT DESCRIPTION

System Overview

The Heart Disease Analysis project focuses on exploring healthcare data related to cardiovascular diseases using modern data analytics and visualization techniques. The system collects health-related data such as patient demographics, lifestyle habits, and medical indicators, and stores it in a structured database. Using SQL queries, the stored data is analyzed and processed to identify meaningful patterns. The processed information is then visualized through interactive dashboards created using Tableau. Finally, the dashboards are embedded into a web interface developed using Flask so that users can interact with the visual insights easily through a browser.

Problem Domain

Heart disease continues to be one of the leading causes of death globally. Several factors such as sedentary lifestyle, smoking habits, unhealthy diet, obesity, and mental stress contribute to an increased risk of cardiovascular diseases. Healthcare institutions collect vast amounts of patient data, but extracting useful insights from this raw data is difficult without analytical tools. This project aims to convert raw healthcare data into visual insights that help in understanding the relationship between lifestyle indicators and heart disease risks.

Core Technology Used

- SQL – used for database storage, querying, and data analysis
- Tableau – used for creating interactive dashboards and visualizations
- Python Flask – used to integrate the dashboards into a web interface
- CSV dataset – used as the primary data source
- GitHub – used for version control and project collaboration

Key Capabilities

- Data exploration using structured SQL queries
- Creation of interactive and dynamic visual dashboards
- Analysis of lifestyle factors such as smoking, BMI, and physical activity
- Identification of patterns and correlations in health data
- Web integration of visual dashboards for easy access

Target Users

- Healthcare professionals and doctors
- Public health researchers and analysts
- Government healthcare departments
- Students learning data analytics and visualization

Real-world Relevance

The insights generated through this project can help healthcare professionals identify high-risk groups and focus on preventive healthcare strategies. Policymakers can also use such analytical dashboards to design awareness campaigns and public health programs aimed at reducing cardiovascular diseases.

3. APPLICATION SCENARIOS / USE CASES

Scenario 1 – Hospital Analysis
A cardiologist working at a hospital can analyze patient information using the dashboard. By examining variables such as BMI, smoking habits, and age groups, the doctor can identify high-risk patients and provide preventive guidance.

Scenario 2 – Government Health Policy
Public health officials can analyze trends across different regions and populations. By studying the relationship between lifestyle indicators and heart disease prevalence, they can recommend policy interventions such as fitness programs or anti-smoking campaigns.

Scenario 3 – Personal Health Monitoring
Individuals with a family history of heart disease can review simplified dashboards provided by healthcare providers to understand their risk factors and adopt healthier habits.

4. TECHNICAL ARCHITECTURE OVERVIEW

The project follows a data analytics pipeline consisting of data collection, database storage, SQL-based data processing, visualization using Tableau dashboards, and web integration using Flask.

Architecture Flow: Data Source → SQL Database → Tableau Visualization → Dashboard → Flask Web Interface → End User

Component Interaction

- Dataset is collected and imported into a SQL database
- SQL queries are executed to explore and analyze the dataset
- Tableau connects directly to the database for visualization
- Interactive dashboards are developed in Tableau
- Dashboards are embedded into a Flask-based web interface

5. PREREQUISITES

Software Requirements

- MySQL Database Server
- MySQL Workbench
- Tableau Desktop
- Python
- Flask Framework
- Visual Studio Code or any IDE

Hardware Requirements

- Minimum 4 GB RAM
- Dual core processor
- System capable of running Tableau and database server

6. PRIOR KNOWLEDGE REQUIRED

- Basic SQL and database management
- Fundamentals of data visualization
- Python programming basics
- Understanding of Tableau dashboards
- Basic web development concepts

7. PROJECT OBJECTIVES

- Analyze heart disease data using SQL queries
- Develop interactive dashboards using Tableau
- Integrate dashboards into a web application using Flask
- Provide insights into lifestyle factors affecting heart disease

8. SYSTEM WORKFLOW

- Collect heart disease dataset
- Store dataset in SQL database
- Perform SQL queries for data exploration
- Connect Tableau with database
- Create visualizations and dashboards
- Embed dashboards into Flask web interface
- Allow users to interact with dashboards

9. MILESTONE 1: REQUIREMENT ANALYSIS & SYSTEM DESIGN

The project begins with identifying the problem of analyzing healthcare data related to heart disease and determining the requirements needed to build the analytical system.

- Define the analytical objectives
- Identify the required technologies
- Design the system architecture
- Plan the database structure

10. MILESTONE 2: ENVIRONMENT SETUP

- Install MySQL database and import dataset
- Install Tableau Desktop for visualization
- Install Python and Flask for web integration
- Create project repository and folder structure

11. MILESTONE 3: CORE SYSTEM DEVELOPMENT

- Dataset imported into database tables
- SQL queries developed for analysis
- Tableau dashboards designed to visualize patterns

12. MILESTONE 4: INTEGRATION & OPTIMIZATION

- Tableau dashboards embedded in Flask web interface
- Optimization of SQL queries for better performance
- Validation of dashboard functionality

13. MILESTONE 5: TESTING & VALIDATION

The system was tested by verifying data loading, SQL query execution, dashboard visualization, and web integration. Each module was tested to ensure correct output and smooth interaction.

14. DEPLOYMENT

- Run Flask web application
- Load Tableau dashboard
- Access dashboards through browser

15. PROJECT STRUCTURE

Heart-Disease-Analysis/
data/
sql/
tableau/
web/
documentation/
README.md

16. RESULTS

The analysis revealed several patterns in the dataset. Higher BMI values and smoking habits show stronger association with heart disease. Age groups above middle age display higher risk levels. Visualization dashboards helped clearly demonstrate these relationships.

17. ADVANTAGES & LIMITATIONS

Advantages

- Interactive visualization of healthcare data
- Easy understanding of complex datasets
- Supports data-driven healthcare insights

Limitations

- Limited dataset size
- No real-time patient monitoring
- No predictive machine learning model

18. FUTURE ENHANCEMENTS

- Integration of machine learning models for prediction
- Cloud-based deployment
- Mobile-friendly dashboard access
- Real-time health data monitoring

19. CONCLUSION

This project demonstrates how SQL, Tableau, and Flask can be combined to analyze healthcare datasets and produce meaningful insights. The developed dashboards help visualize risk factors associated with heart disease and support better data-driven healthcare awareness.

20. APPENDIX

Dataset link:

https://drive.google.com/file/d/190Qmq27LeZZ_nWricP3Obl7ys_5otEsp/view

GitHub Repository:

https://github.com/asleshaxa/Heart-Disease-Analysis

Live Demo Link:

https://heart-disease-analysis-m423.onrender.com/